VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



MATHEMATICAL MODELING (CO2011)

Assignment

"Dynamics of Love"

Instructor(s): Mai Xuân Toàn

Students: Trần Anh Khoa - 2113773 *(Group L04 - Team 02, **Leader**)*
Trần Quốc Thắng - 2110551 *(Group L03 - Team 02)*
Lê Nguyên Chương - 2112945 *(Group L04 - Team 02)*
Bùi Tiến Dũng - 2113055 *(Group L04 - Team 02)*

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Contents

List of Symbols **2**

List of Acronyms **2**

List of Figures **4**

List of Tables **4**

Member list & Workload **4**

1 Section 1 **5**

1.1 Subsubsection 1 5

List of Symbols

$\mathbb{N}$ Set of natural numbers

$\mathbb{R}$ Set of real numbers

$\mathbb{R}^+$ Set of positive real numbers

List of Acronyms

ODE (First-Order) Ordinary Differential Equation

IVP Initial-Value Problem

LTE Local Truncation Error

DS Dynamical System

Fig. Figure

Tab. Table

Sys. System of Equations

Eq. Equation

e.g. For Example

i.e. That Is

List of Figures

1 Red-black tree 5

List of Tables

1 Member list & workload 4

Member list & Workload

No.	Fullname	Student ID	Problems	% done
1	Trần Anh Khoa	2113773	- Exercise 1: 1.2 - Exercise 2 - Exercise 3: 3.2	100%
2	Trần Quốc Thắng	2110551	- Exercise 1: 1.3 - Exercise 2 - Exercise 3: 3.1 - Exercise 4	100%
3	Lê Nguyên Chương	2112945	- Exercise 1: 1.1 - Exercise 4 - LaTeX	100%
4	Bùi Tiến Dũng	2113055	- Exercise 1 - Exercise 3: 3.1 - Exercise 4	100%

Table 1: Member list & workload

1 Section 1

1.1 Subsubsection 1



Figure 1: Red-black tree

```python
import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int)  #dummy variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

                if M is None:
                    M = np.copy(VT)
                else:
                    M = np.concatenate((M, VT), 1)

```

```
                VT = np.zeros((n*m,1), int)

    return M
```

Listing 1: Python example